Description of the PennyLane Resource Estimation Experiment

P765 Final Project

March 31, 2026

1 Purpose

This report documents the resource-estimation workflow used for the already-selected subspace basis states in the project. The comparison covers three basis-state families:

- Krylov basis states,
- UCCSD basis states,
- fixed-architecture hardware-efficient ansatz (HEA) basis states,

for the molecules

$$H_2, \quad H_4, \quad LiH.$$

The goal is not to rerun the subspace optimization. Instead, the goal is to estimate the resource cost of the basis states that were already selected in the validated subspace benchmark notebooks.

2 Method

The experiment is implemented in `Resource_Estimation_Subspace.ipynb`. It uses two PennyLane resource layers:

- `qml.specs(...)` for native circuit metrics such as gate count, depth, and two-qubit gate count,
- `pennylane.estimator.estimate(...)` for logical gate-set estimates.

The notebook uses *frozen selected basis-state specifications* rather than rerunning the subspace optimization. This means the resource report is tied directly to the selected basis states that were already validated in the benchmark workflow.

For the native-resource view, the notebook evaluates state-preparation circuits on `null.qubit` and records:

- total native gate count,
- circuit depth,
- gate-type counts,
- gate-size counts,
- two-qubit gate count.

For the logical-resource view, the notebook uses PennyLane's default logical gate set and records:

- total wires,
- algorithmic wires,
- allocated wires in the zero or unknown state,
- total logical gate count,
- logical gate-count breakdown.

Prep-only subspace proxy

The report also includes a prep-only proxy for the total subspace-method cost. For a basis of size K:

- each selected basis state contributes its own state-preparation resources,
- each unique off-diagonal overlap entry S_{ij} contributes a proxy cost of $G_i + G_j$,
- each unique off-diagonal Hamiltonian entry H_{ij} contributes a proxy cost of $G_i + G_j$,
- each diagonal Hamiltonian entry H_{ii} contributes a proxy cost of G_i.

This is intentionally a *prep-only proxy*. It is not a full Hadamard-test model and not a shot-complexity model. Its purpose is to compare how the resource cost grows as the number of selected basis states increases.

Important H_4 note

For H_4, the frozen Krylov selected times are exact. The H_4 UCCSD and fixed-HEA entries use same-shape representative angle vectors together with the validated selected-basis availability from the benchmark sweep. This is acceptable for the present report because the PennyLane resource counts used here depend on the circuit structure, not on the specific continuous angle values.

3 Systems and Frozen Inputs

All three systems use PennyLane quantum chemistry Hamiltonians in the `sto-3g` basis with coordinates interpreted in `angstrom`.

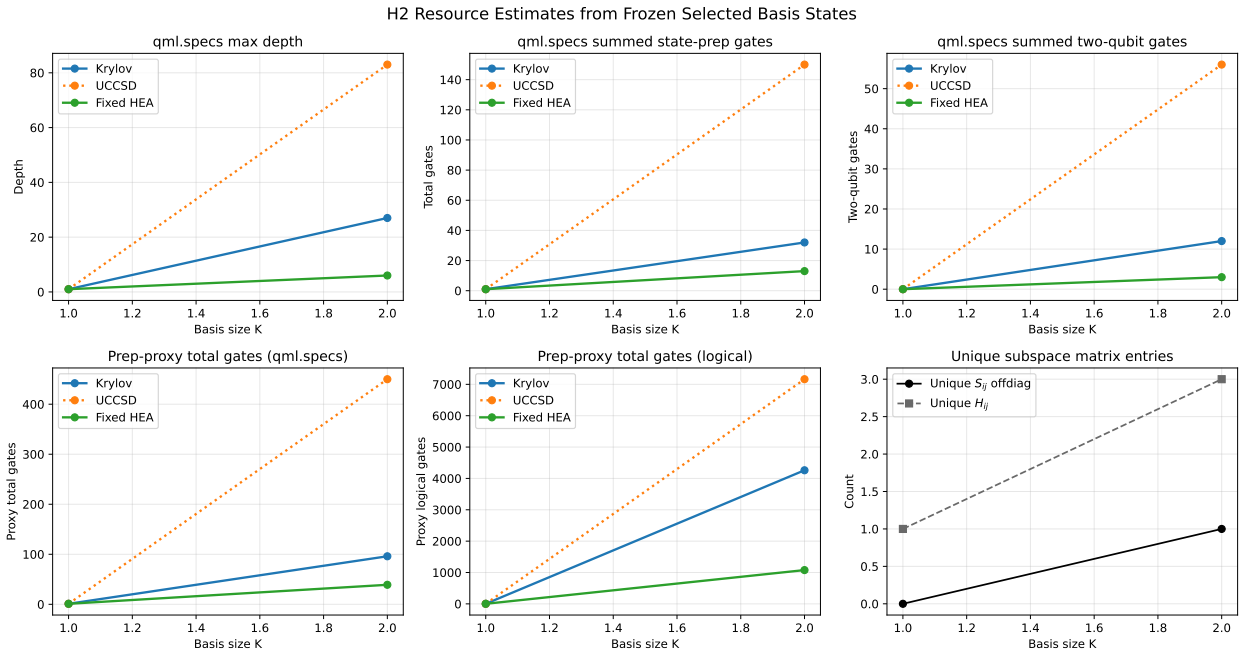
Molecule	Bond length (Å)	Basis	Qubits	Electrons	Pauli terms	Available frozen K
H_2	0.74	`sto-3g`	4	2	15	all families through $K = 2$
H_4	0.74	`sto-3g`	8	4	185	Krylov through $K = 3$, UCCSD/HEA through $K = 6$
LiH	1.5474	`sto-3g`	12	4	631	all families through $K = 3$

4 Results

Each table below is populated from the validated `subspace_proxy_rows` output of the notebook after a headless run with plot export enabled.

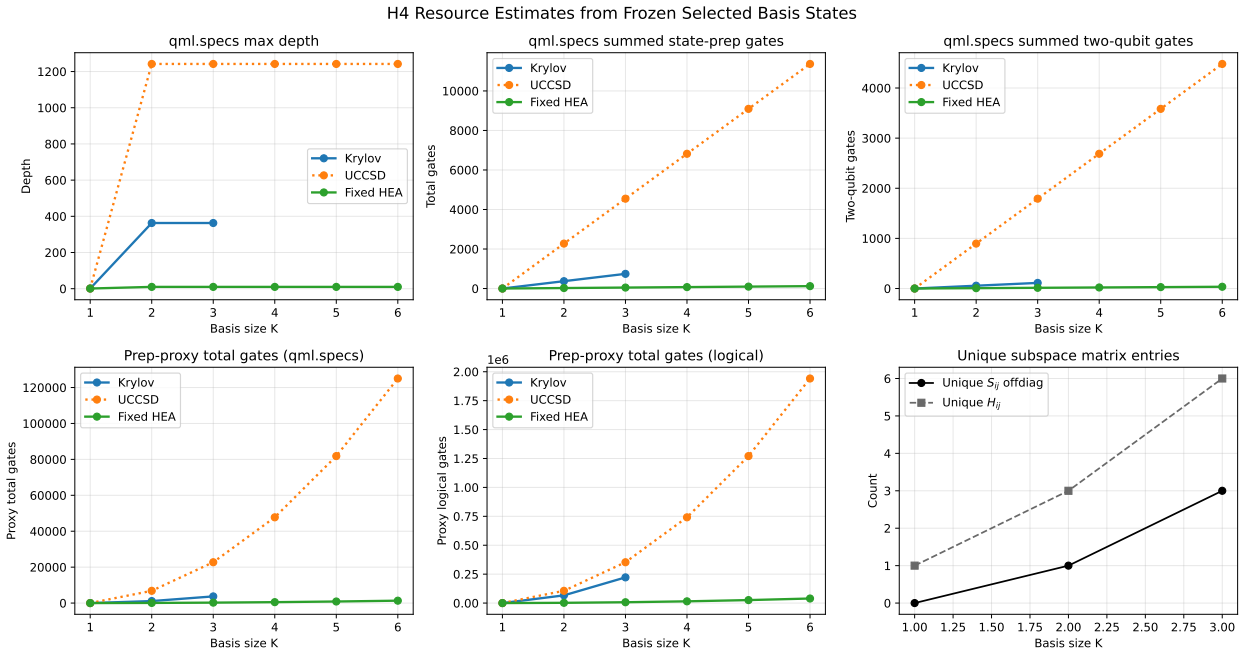
H_2

Family	K	prep_sum_num_gates	prep_max_depth	prep_sum_two_qubit_gates	specs_proxy_total_gates	logical_proxy_total_gates	unique (S_{ij}, H_{ij})
Krylov	1	1	1	0	1	2	$(0,1)$
Krylov	2	32	27	12	96	4260	$(1,3)$
UCCSD	1	1	1	0	1	2	$(0,1)$
UCCSD	2	150	83	56	450	7164	$(1,3)$
Fixed HEA	1	1	1	0	1	2	$(0,1)$
Fixed HEA	2	13	6	3	39	1077	$(1,3)$



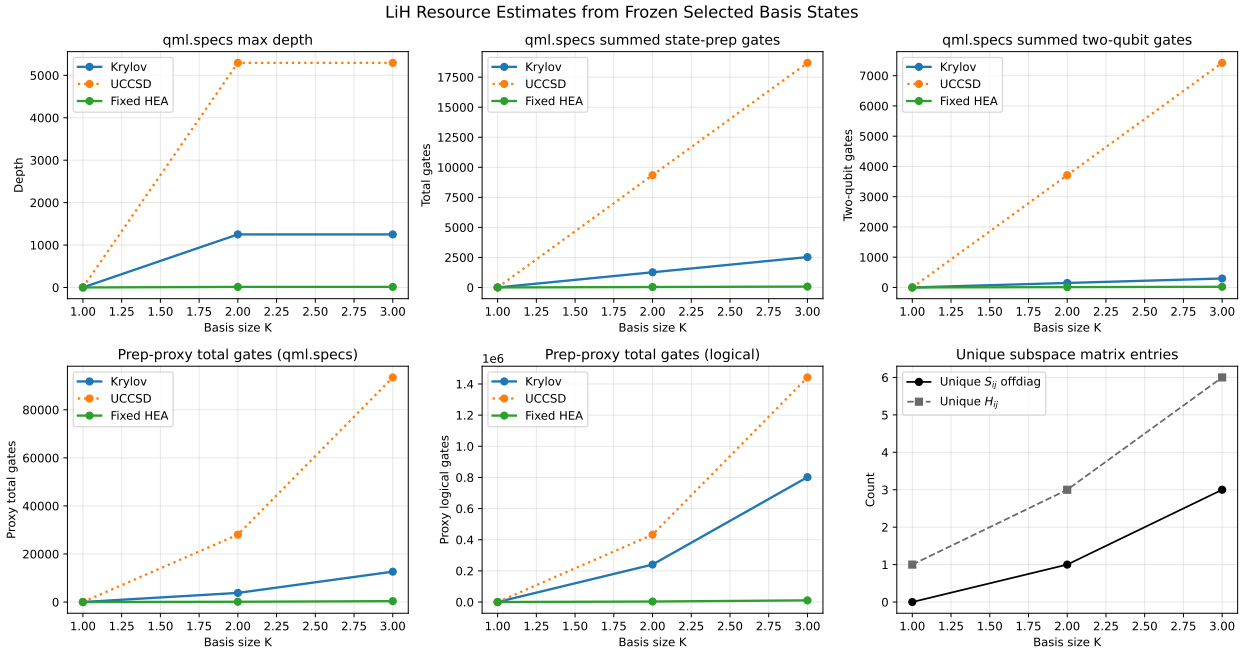
H_4

Family	K	prep_sum_num_gates	prep_max_depth	prep_sum_two_qubit_gates	specs_proxy_total_gates	logical_proxy_total_gates	unique (S_{ij}, H_{ij})
Krylov	1	1	1	0	1	4	$(0,1)$
Krylov	2	372	363	56	1116	66648	$(1,3)$
Krylov	3	743	363	112	3715	222140	$(3,6)$
UCCSD	1	1	1	0	1	4	$(0,1)$
UCCSD	2	2274	1242	896	6822	105912	$(1,3)$
UCCSD	3	4547	1242	1792	22735	353020	$(3,6)$
UCCSD	4	6820	1242	2688	47740	741328	$(6,10)$
UCCSD	5	9093	1242	3584	81837	1270836	$(10,15)$
UCCSD	6	11366	1242	4480	125026	1941544	$(15,21)$
Fixed HEA	1	1	1	0	1	4	$(0,1)$
Fixed HEA	2	25	10	7	75	2157	$(1,3)$
Fixed HEA	3	49	10	14	245	7170	$(3,6)$
Fixed HEA	4	73	10	21	511	15043	$(6,10)$
Fixed HEA	5	97	10	28	873	25776	$(10,15)$
Fixed HEA	6	121	10	35	1331	39369	$(15,21)$



LiH

Family	K	prep_sum_num_gates	prep_max_depth	prep_sum_two_qubit_gates	specs_proxy_total_gates	logical_proxy_total_gates	unique (S_{ij}, H_{ij})
Krylov	1	1	1	0	1	4	$(0,1)$
Krylov	2	1264	1251	148	3792	240528	$(1,3)$
Krylov	3	2527	1251	296	12635	801740	$(3,6)$
UCCSD	1	1	1	0	1	4	$(0,1)$
UCCSD	2	9346	5294	3712	28038	432600	$(1,3)$
UCCSD	3	18691	5294	7424	93455	1441980	$(3,6)$
Fixed HEA	1	1	1	0	1	4	$(0,1)$
Fixed HEA	2	37	14	11	111	3225	$(1,3)$
Fixed HEA	3	73	14	22	365	10730	$(3,6)$



5 Interpretation

Several consistent trends appear across the three systems:

- Krylov uses the smallest basis-state parameterization, but its Trotterized evolution can still accumulate a large logical proxy cost as the system grows.
- UCCSD has the largest native and logical costs for the larger systems, especially for LiH, where both the depth and logical proxy totals become very large even at modest K.
- The fixed HEA remains much shallower and lighter under the current one-layer architecture, even though its subspace-quality improvement was more limited in the energy benchmarks.
- The prep-only subspace proxy grows quickly with K because the number of unique overlap and Hamiltonian matrix elements grows quadratically:

$$\#S_{\text{offdiag}} = \frac{K(K-1)}{2}, \qquad \#H_{\text{unique}} = \frac{K(K+1)}{2}.$$

In other words, this report makes two costs visible at the same time:

- the cost of preparing each selected basis state,
- the rapid growth in total subspace-evaluation work as the basis size increases.

6 Implementation Files

This resource-estimation package is implemented in:

- `Resource_Estimation_Subspace.ipynb`
- `Resource_Estimation_Subspace_Description.tex`